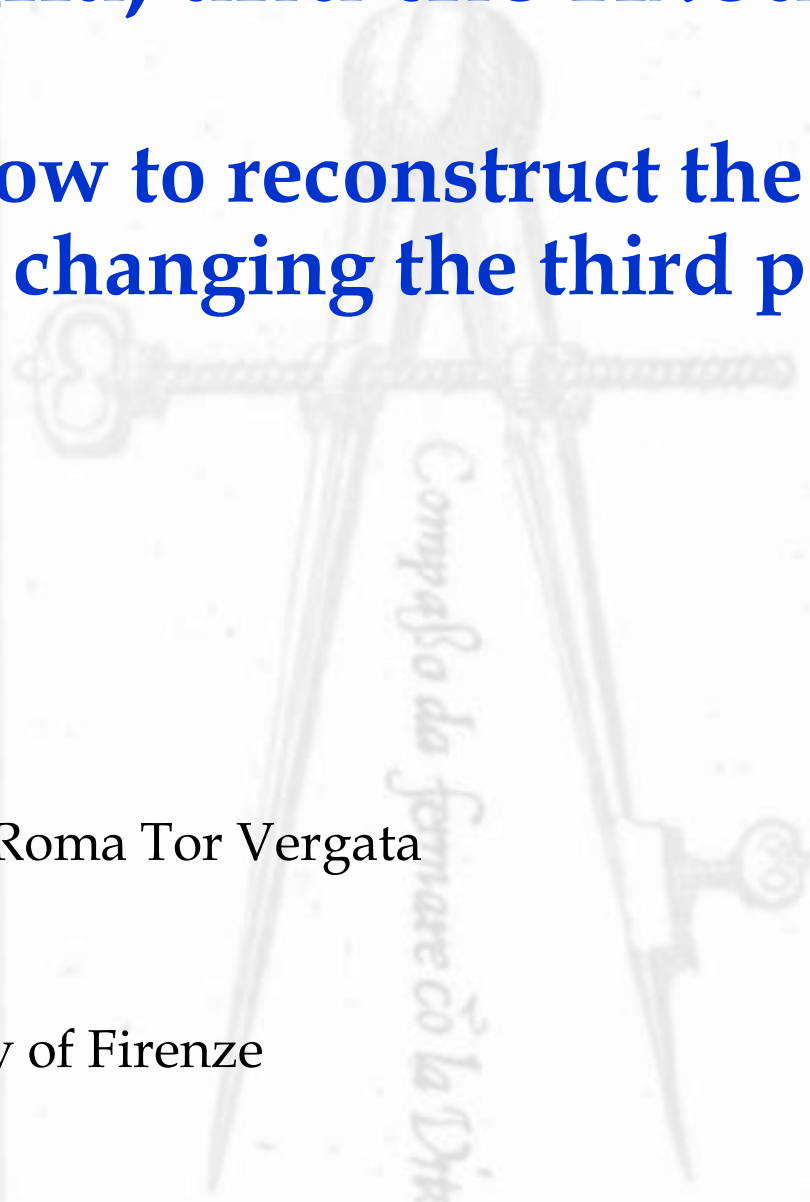


Ferrari, Tartaglia, and the fixed-opening compass

or how to reconstruct the *Elements*
by changing the third postulate



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20-24 July
2026
Aveiro, Portugal



European Summer University on History
and Epistemology in Mathematics Education

Aveiro, Portugal

20-24 july 2026

The agenda

- Historical and mathematical context
- Educational goals of the activity (designed for a high school class).
- First activity (and feedback)
- Second activity
- How the story ended and final discussion

10th European Summer University
History and Epistemology in Mathematics Educ

1539: Girolamo Cardano was writing a manual on practical arithmetic called the *Practica Arithmeticae*, which he hoped would be commercially successful.

This was no easy task, as the standard reference work was Luca Pacioli's *Summa*, an encyclopedic work that had been very successful.

In the *Summa*, Pacioli argued that a general formula for solving cubic equations had not yet been found but that certain specific cases could be solved.



Cardano learned that an unknown abacus teacher named Tartaglia had sensationally defeated his opponent in a mathematical dispute centered on solving cubic equations a few years earlier.

Cardano convinced Tartaglia to reveal the solution formula, on the condition that Cardano not publish it in his treatise.

1542: during a trip to Bologna, Cardano and his student Ferrari discovered that the mathematician Scipione del Ferro had already found the third-degree equation solving formula in the early 16th century.

Cardano therefore considered himself released from his oath and, in **1545**, published the *Ars Magna*, which included formulas for solving cubic and quartic equations.



1546 *Quesiti et inventioni diverse*. Tartaglia published his correspondence with the aim of exposing Cardano's improper conduct.

Cardano did not respond personally. Instead, his student Ludovico Ferrari challenged Tartaglia to a public mathematical duel.

QVESITI ET INVEN-
TIONI DIVERSE
DE NICOLO TARTAGLIA,
DI NOVO RESTAMPATI CON VNA
GIONTA AL SBSTO LIBRO, NELLA
quale si mostra duoi modi di redur una Città inespugnabile.
LA DIVISIONE ET CONTINENTIA DI TUTTA
l'opra nel seguente foglio si trouara notata.
CON PRIVILEGIO



LIBRO NONO DELL
QVESITI, ET INVENTIONI DIVERSE,
DE NICOLO TARTAGLIA.

Sopra la scientia Arithmetica, Geometrica, & in la Pratica Speculatiua
de Algebra, & Almucabala, uolgarmente detta Regola de
la cosa, ouer Arte maggiore, & massime della
inuentione de Capitoli de Cosa, e Cubo
equal à numero, & altri suoi
ederenti, et dependenti,

Et similmente de cens, e cubi equal à numero, & suoi
dependenti, quali dalli Sapienti sono stati
giudicati impossibili.

I SEI CARTELLI

DI MATEMATICA DISFIDA

PRIMAMENTE INTORNO ALLA GENERALE RISOLUZIONE DELLE EQUAZIONI CUBICHE

DI

LODOVICO FERRARI

COI SEI CONTRO-CARTELLI IN RISPOSTA

DI

NICOLÒ TARTAGLIA,

COMPRENDENTI

LE SOLUZIONI DE' QUESITI DALL' UNA E DALL' ALTRA PARTE PROPOSTI

RACCOLTI, AUTOGRAFATI E PUBBLICATI

DA ENRICO GIORDANI,

Bolognese.

Premesse notizie bibliografiche ed illustrazioni sui Cartelli medesimi,
estratte da documenti già a stampa ed altri
manoscritti favoriti dal Comm. Prof. SILVESTRO GHERARDI, Preside dell'Istit. Tecn. Prov. di Firenze.



MILANO, 1876.

Tartaglia proposed 31 questions to Ferrari. Some of these questions were about geometry and required the reconstruction of certain geometrical proofs using a fixed compass and a ruler instead of a ruler and a usual compass.

In other words, Tartaglia was asking for a variation of the third postulate.

Translation of the *Elements*, edited by Tartaglia.
(1 ed. 1543)

Petitione.iii.

² Anchora adimandamo che ce sia concesso che sopra a qualunque
³ centro ne piace puotemo designare uno cerchio di che grandezza ci pare.

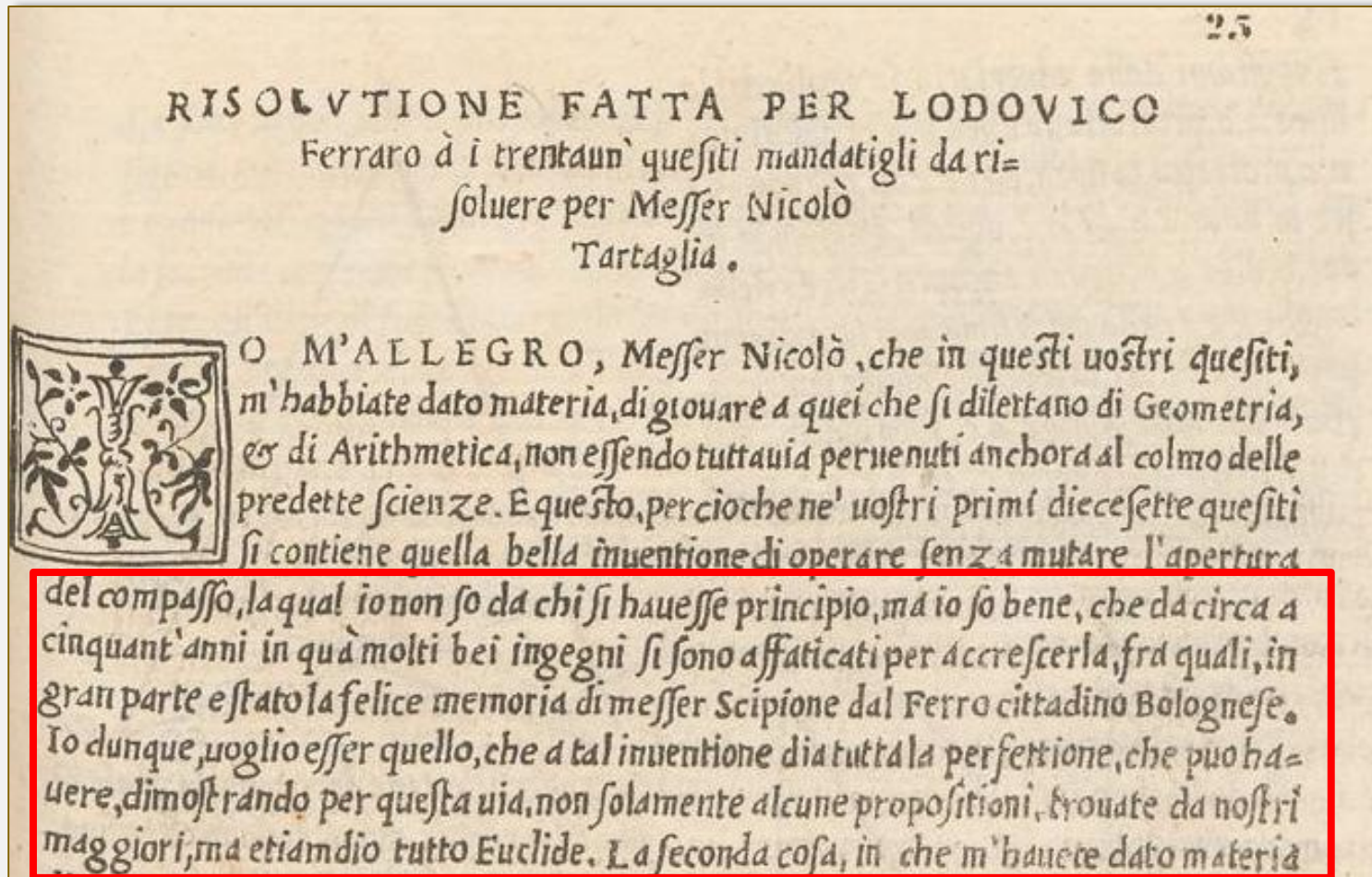
Sopra Euclide.

E Glie manifesto, Euclide Megarense non solamente esser el primo, (Mala guida, & scorta) de tutti quelli che delle Discipline Matematiche hanno trattato, e per tanto, me aparso primamente di proporreui alcuni suoi problemiche quel ne insegna di concludere geometricamente dimostratiuamente, Giongendoui solamente questa sotilita, che cadauno de quelli sia concluso cō qual si voglia apertura di compasso proposta dal Auerfario, cioe senza mai mouere lo detto compasso di tal data appertura con atti, & regole generale dimostratiue, cioe concedendoui tutte le sue Pettitioni & commune sententie del detto Euclide eccetto la sua seconda, ouer terza petitione, cioe quella doue che adimanda che gli sia concesso che sopra a qualunque centro che gli pare di poterui designare vn cerchio di che grandezza gli pare, Ma in luoco di quella vi pongo quest'altra: cioe che sopra a qual si voglia centro ve pare vi concedo che gli possa designare vn cerchio secondo la quantita della data apertura di compasso, cioe proposta dal auersario, secondo che a lui pare (pur che non sia in retta linea) Hor per dar principio incominceremo dalle cose piu facile secondo l'ordine de naturali.



Euclid	Tartaglia and Ferrari
Let the following be postulated: To draw a straight line from any point to any point.	
To produce a finite straight line continuously in a straight line.	
To describe a circle with any center and radius.	To describe a circle with any center and a fixed radius.
That all right angles equal one another.	
That, if a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.	

Euclidean propositions to be proven using a compass with a fixed opening: III.17, VI.25, VI.28, VI.29, X.31, X.32, X.33, XIII.18, tangents to conics ... (17 problems)



Tartaglia asked Ferrari to prove a few specific propositions using only a straightedge and a compass with a fixed opening. **However, Ferrari replied that he had proven «the whole of Euclid»**

Let's clarify what Ferrari means by this statement.

- If it is necessary to construct circles with a radius different from the assigned one, it suffices to determine three distinct points through which the circle passes to consider it constructed.
- Not all of Euclid's propositions in the *Elements* are re-demonstrate; rather, the logical dependence among the propositions is studied, so that only the group of propositions on which the others depend need be proven.

Your tasks

1. Ferrari's reconstruction of the *Elements* does not follow the Euclidean numbering system.

What will Ferrari's starting proposition be for «reconstructing» the *Elements*?

Participants must study the logical dependence of the first (eleven) propositions of the *Elements* in order to evaluate and rule out those that depend directly or indirectly on the third Euclidean postulate.

2. Once the first proposition of the Ferrari-*Elements* has been determined and agreed upon, **participants will be asked to prove some subsequent propositions using only a fixed compass.**

To answer this question, participants must develop a proof that considers the imposed constraints (the fixed compass) and the logical hierarchy. An object such as the *midpoint* or *bisector* will only be available once it has been constructed with the fixed compass. This activity can be carried out with a fixed compass or, alternatively, in a dynamic geometry environment with an appropriate macro (we will see this later).

HOW TO CONDUCT THE WORKSHOP IN THE CLASSROOM

First option: Students are presented with the geometric problem, «Prove certain propositions using a fixed-opening compass and a ruler». Later the problem is placed in its historical context, so to give students the opportunity to study Ferrari's solutions by reading the sources in the original language or translation and compare them with their own solutions. This option was tested in a third-year class at a scientific high school in Florence.

Second option: The activity begins with the description of the historical context. Students are asked to put themselves in Ferrari's shoes and answer his questions. As in the first option, part of the activity is dedicated to studying the historical sources.

In both cases, the activities are carried out in groups.

Goals:

- Create (rather than reproduce) proofs.
- Develop or strengthen argumentative skills through group work.
- Develop an understanding of what a logical-deductive structure means, as well as how the theory derived from it depends on the choice of postulates/axioms.
- Understand the meaning of «construction with a straightedge and (a fixed opening) compass

What will Ferrari's starting point be for «reconstructing» the *Elements*?

Maybe the first proposition?

Let's take a quick look at it.

Proposition 1. Problem 1

To construct an equilateral triangle on a given finite straight line.

Let AB be the given finite straight line.

It is required to construct an equilateral triangle on the straight line AB .

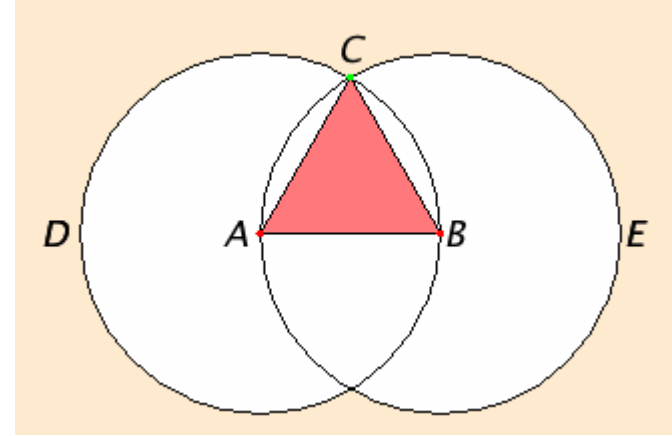
Describe the circle BCD with center A and radius AB . Again describe the circle ACE with center B and radius BA . Join the straight lines CA and CB from the point C at which the circles cut one another to the points A and B .

Now, since the point A is the center of the circle CDB , therefore AC equals AB . Again, since the point B is the center of the circle CAE , therefore BC equals BA . But AC was proved equal to AB , therefore each of the straight lines AC and BC equals AB .

And things which equal the same thing also equal one another, therefore AC also equals BC .

Therefore the three straight lines AC , AB , and BC equal one another.

Therefore the triangle ABC is equilateral, and it has been constructed on the given finite straight line AB .



Postulate III

Postulate I

Def.15

This proposition is only acceptable if the side coincides with the opening of the compass

Comm. Not. 1

Def.20

Your first task is to find, among the first ten propositions of the *Elements*, one that does not depend on the Third Postulate or on propositions that depend on the Third Postulate.

Does such a proposition exist?

[Tinyurl.com/fixed-comp](https://tinyurl.com/fixed-comp)



Livre I

	Df	Dem	N. C.	Prop.
Prop. 1	15, 20	1, 3	1	
Prop. 2	15, 20	1, 2, 3	1, 3	1
Prop. 3	15	3	1	2
Prop. 4			7, "9"	
Prop. 5		1, 2	3	3, 4
Prop. 6		1	8	3, 4
Prop. 7		1	8	5
Prop. 8			7	7
Prop. 9	20	1		
Prop. 10	20			
Prop. 11	10, 20	1		
Prop. 12	10, 15	1, 3		
Prop. 13	10		1, 2	
Prop. 14		2, 4	1, 2, 3, 8	
Prop. 15		4	1, 2, 3	
Prop. 16		1, 2	8	
Prop. 17		2	"4"	
Prop. 18		1	8	
Prop. 19				
Prop. 20		1, 2	8	
Prop. 21		2	"4"	
Prop. 22	15	1, 3	1	
Prop. 23		1		
Prop. 24		1	1, 8	
Prop. 25				
Prop. 26		1	1, 8	
Prop. 27	23	2		
Prop. 28		4	1, 2, 3	
Prop. 29	23	2, 5	1, 2, "4"	
Prop. 30			1	
Prop. 31		1, 2		
Prop. 32		2	1, 2	
Prop. 33		1		
Prop. 34		1	2	
Prop. 35			1, 2, 3	
Prop. 36		1	1	
Prop. 37		2	"6"	
Prop. 38		2	"6"	
Prop. 39		1	1, 8	
Prop. 40		1	1, 8	
Prop. 41		1	1, 2	
Prop. 42		1	1, 2	
Prop. 43		1	2, 3	
Prop. 44		1, 2, 5	1, 8	
Prop. 45		1	1, 2	
Prop. 46	22	4	1, 3	
Prop. 47		1, 4	1, 2, "5"	
Prop. 48		1	1, 2	

	Df	Dem	N. C.	Prop.
Prop. 1	15, 20	1, 3	1	
Prop. 2	15, 20	1, 2, 3	1, 3	1
Prop. 3	15	3	1	2
Prop. 4			7, "9"	
Prop. 5		1, 2	3	3, 4
Prop. 6		1	8	3, 4
Prop. 7		1	8	5
Prop. 8			7	7
Prop. 9	20	1		1, 3, 8
Prop. 10	20			1, 4, 9
Prop. 11	10, 20	1		1, 2-3, 8
Prop. 12	10, 15	1, 3		8, 10
Prop. 13	10		1, 2	11
Prop. 14		2, 4	1, 2, 3, 8	13
Prop. 15		4	1, 2, 3	13
Prop. 16		1, 2	8	2-3, 4, 10, 15

The first proposition that does not depend, either directly or indirectly, on Postulate III is Proposition I.4, known today as the Side-Angle-Side (SAS) condition

A photograph of a manuscript page from Simon Stevin's 'De Equilibrio' (1586), showing Proposition I.4 in Italian. The text is written in a cursive script and reads: "Prima propositione, ch'io piglio da dimostrare, si è la quarta del primo, laquale io dimostro in punto com'il Theone, atteso che la non ha bisogno delle propositioni antecedenti, ne anche della petitione eccettuata."

The first proposition I am considering is Proposition I.4. I prove it in accordance with Theon's version. This proposition does not depend on the preceding propositions or the eliminated postulate.

Proposition I.4

If two triangles have two sides equal to two sides respectively, and have the angles contained by the equal straight lines equal, then they also have the base equal to the base, the triangle equals the triangle, and the remaining angles equal the remaining angles respectively, namely those opposite the equal sides.

F.1	<i>Elements</i> I.4	If two triangles have two sides equal to two sides respectively, and have the angles contained by the equal straight lines equal, then they also have the base equal to the base, the triangle equals the triangle, and the remaining angles equal the remaining angles respectively, namely those opposite the equal sides.
F.2	<i>Elements</i> I.5	In isosceles triangles the angles at the base equal one another, and, if the equal straight lines are produced further, then the angles under the base equal one another.
F.3	<i>Elements</i> I.8	If two triangles have the two sides equal to two sides respectively, and also have the base equal to the base, then they also have the angles equal which are contained by the equal straight lines.
F.4	<i>Elements</i> I.9	To bisect a given rectilinear angle.
F.5	<i>Elements</i> I.10	To bisect a given finite straight line.
F.6	<i>Elements</i> I.11	To draw a straight line at right angles to a given straight line from a given point on it.

Your second task is to prove Propositions I.5, I.8, I.9, I.10 and I.11 using a fixed compass, while also respecting the logical dependence between the propositions.

If you would like to see Ferrari's proofs, you can find them here: [Tinyurl.com/fixed-comp](https://tinyurl.com/fixed-comp)



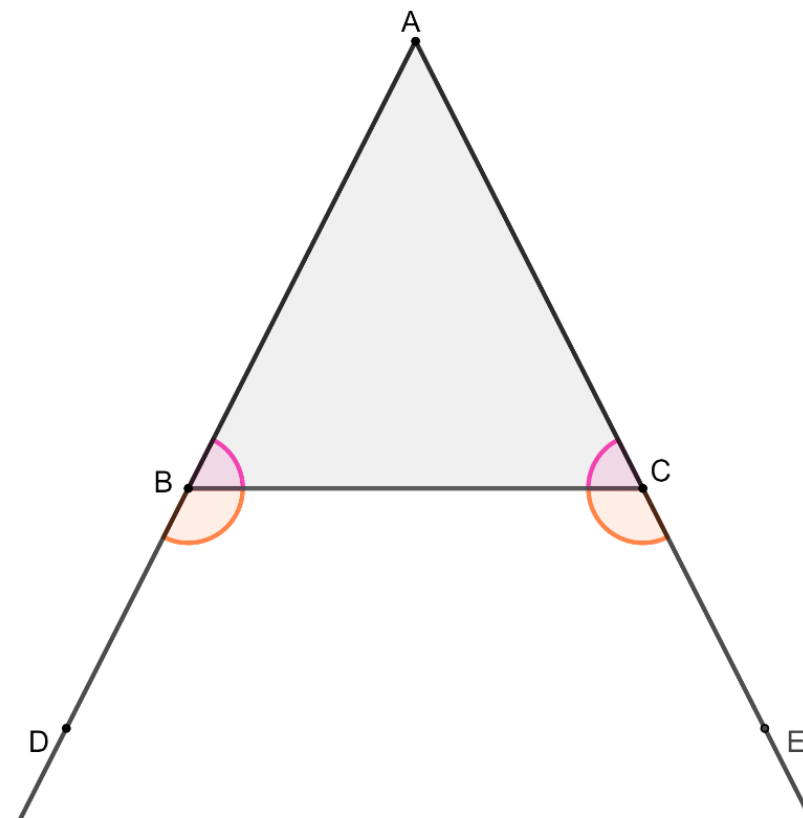
How to Build the “Fixed Compass” Macro in GeoGebra

- Using **Circonferenza: centro e raggio** (Circle: Center and Radius) tool to draw a circle with an arbitrary radius.
- Go to the top menu and select **Strumenti → Crea Nuovo Strumento** ('Tools' → 'Create New Tool')
- In the **Oggetti finali** (Final Objects) tab, select the circle you just created from the drop-down menu.
- In the **Oggetti iniziali** (Initial Objects) tab, only point A should appear.
- In the **Nome e Icona** (Name and Icon) tab, give the tool a name (e.g. Fixed Compass), choose an image if desired and click Fine (Done).

Enjoy your work

The second proposition that Ferrari proves is I.5.

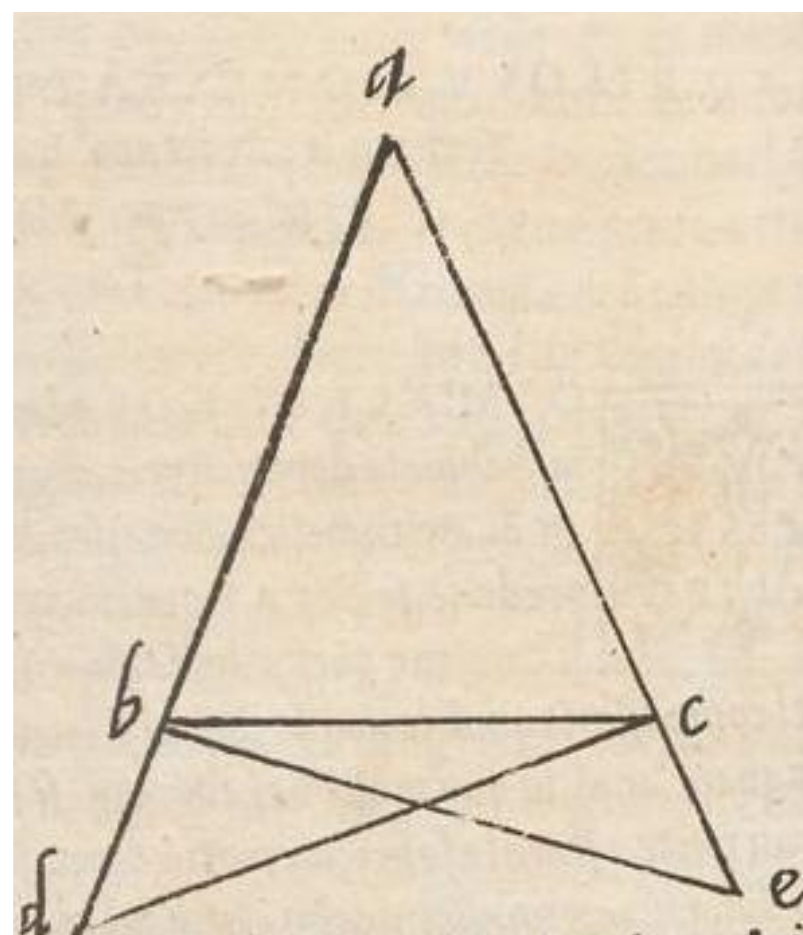
I.5 In isosceles triangles the angles at the base equal one another, and, if the equal straight lines are produced further, then the angles under the base equal one another.

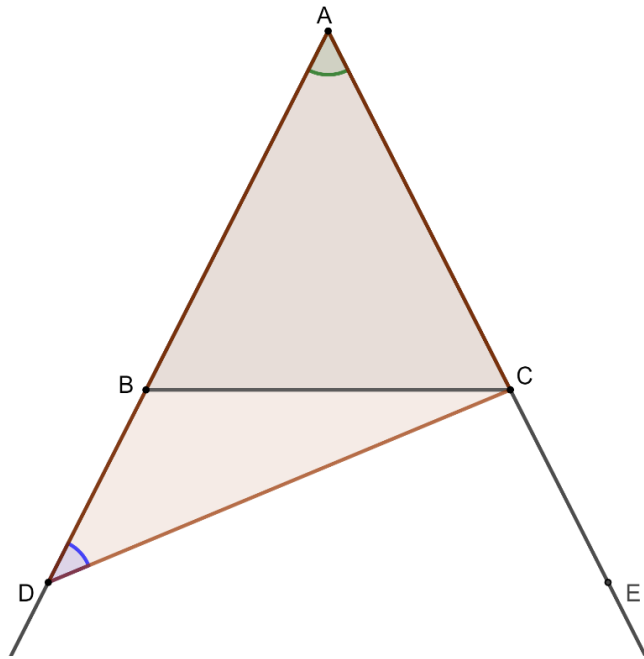


The Euclidean proof requires I.3, which depends on Postulate III. Therefore, Ferrari modifies it.

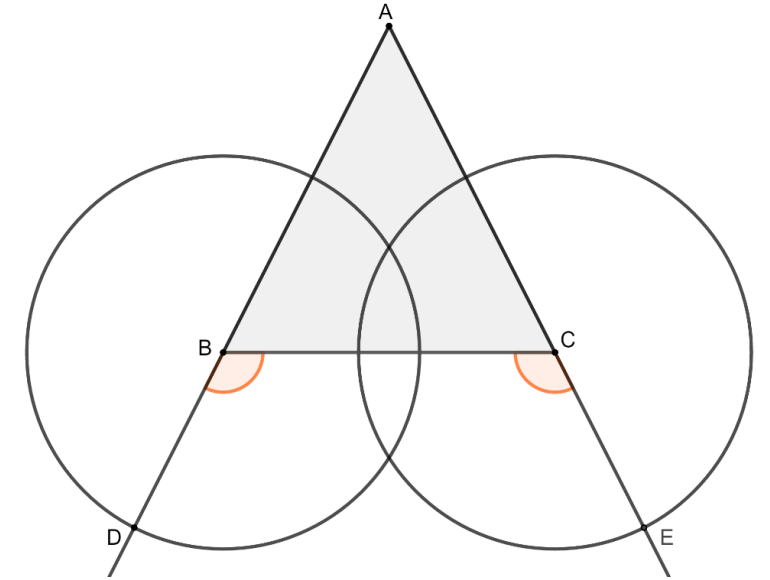
La seconda sarà la quinta del primo, nella quale, perch'io nō mi posso servire della terza, io procedo per questa uia. Sia il Triangolo a, b, c , del quale i due lati, a, b , & a, c , siano uguali insieme. Per la seconda petitione, io tiro le due, a, b , & a, c , in lungo, inde finitamente: poi, facendo per centro i due ponti, b , & c , secondol'apertura del compasso,

propostami dallo aduersario, io taglio dalla linea a, b , protratta, la linea b, d , et dalla linea a, c , protratta, la linea c, e : e' così, essendo poste le linee, a, b , & a, c , uguali, & essendo le due, b, d , & c, e , uguali anchora esse, & della longhezza della medesima apertura di compasso, seguita, per la seconda comune sentēza, che le due linee, a, d , & a, e , sono uguali. Del resto, tiro le due linee, d, c , & b, e , e' compio la dimostratione, come il Theone.



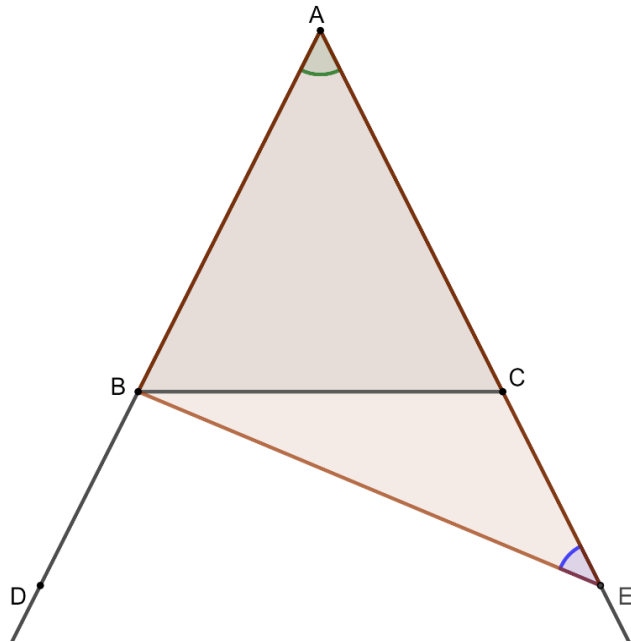


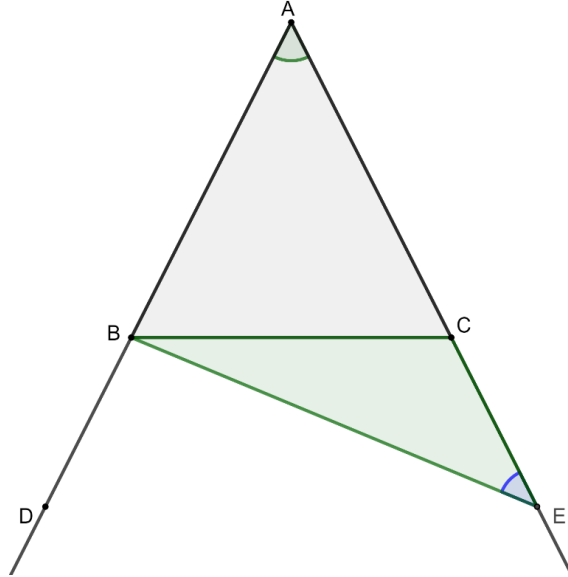
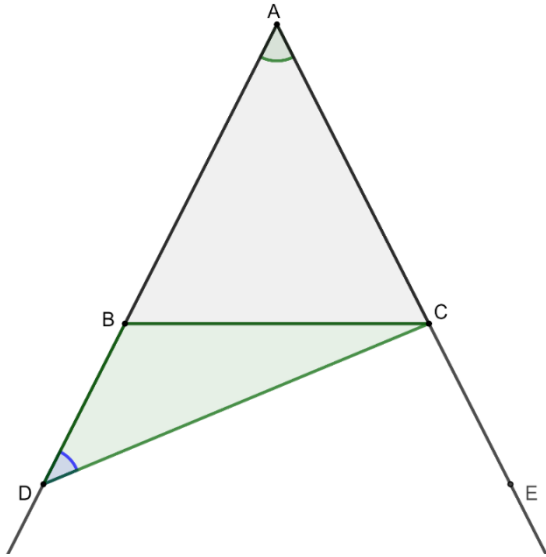
I assume that AB and AC are equal sides, which I can extend indefinitely according to Postulate II.



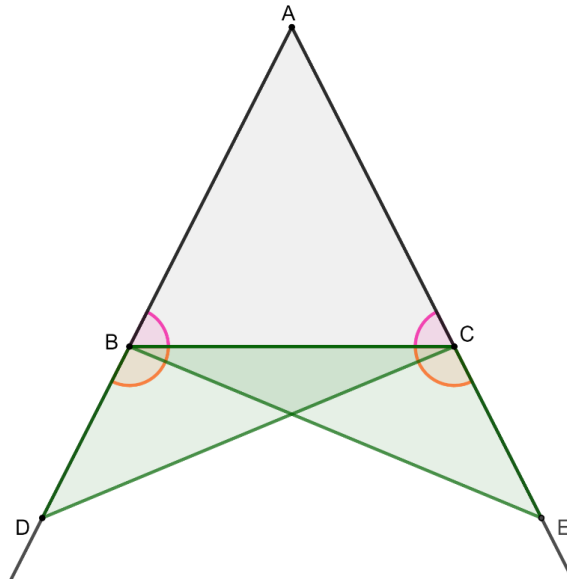
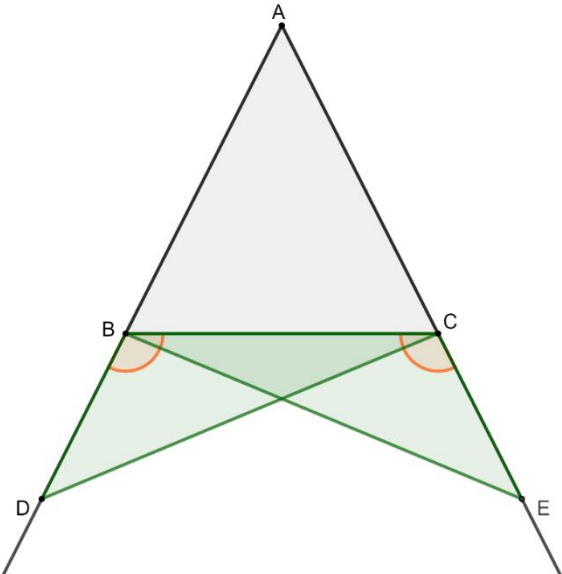
Consider triangles ACD and ABE , which are congruent by the SAS condition (I.4)

Therefore, I conclude that angles ADC and AEB are also congruent.





I consider triangles BDC and ECB, which are congruent by I.4 because BD and EC are congruent differences of line segments (Common Notion III). Therefore, I conclude that the angles CBD and ECB are congruent.



According to Common Notion III, angles ABC and BCA are congruent because they are the difference of two congruent angles (ABE and CBE, and DCA and DCB, respectively).

But doesn't Proposition I.8 rely on the Third Postulate?

	Df	Dem	N. C.	Prop.
Prop. 1	15, 20	1, 3	1	
Prop. 2	15, 20	1, 2, 3	1, 3	1
Prop. 3	15	3	1	2
Prop. 4			7, "9"	
Prop. 5		1, 2	3	3, 4
Prop. 6		1	8	3, 4
Prop. 7		1	8	5
Prop. 8			7	7
Prop. 9	20	1		1, 3, 8
Prop. 10	20			1, 4, 9
Prop. 11	10, 20	1		1, 2-3, 8
Prop. 12	10, 15	1, 3		8, 10
Prop. 13	10		1, 2	11
Prop. 14		2, 4	1, 2, 3, 8	13
Prop. 15		4	1, 2, 3	13
Prop. 16		1, 2	8	2-3, 4, 10, 15
			...	...

I.8 depends on I.7

I.7 depends on I.5

I.5 depends on I.3
and I.4

Proof I.3 depends
on Postulate III.
However, Ferrari
constructed a
proof of I.5 that
does not depend
on Postulate III.

Ferrari then proceeds to prove Proposition I.9.

Proposition I.9

To bisect a given rectilinear angle.

Let's briefly look at the Euclidean proof

F.4

La quarta di queste sarà la nona del primo. Sia dunque l'angolo, che si ha da dividere per mezzo, l'angolo b, a, c : & sian tirate, per la seconda petitione, le linee a, b , & a, c , in lungo indefinitamente. Io facendo centro il

Proposition I.9

To bisect a given rectilinear angle.

Let the angle BAC be the given rectilinear angle.

It is required to bisect it.

Take an arbitrary point D on AB . Cut off AE from AC equal to AD

[Proposition I.3], and join DE [Postulate I]. Construct the equilateral triangle DEF on DE , and join AF [Proposition I.1, Postulate 1]

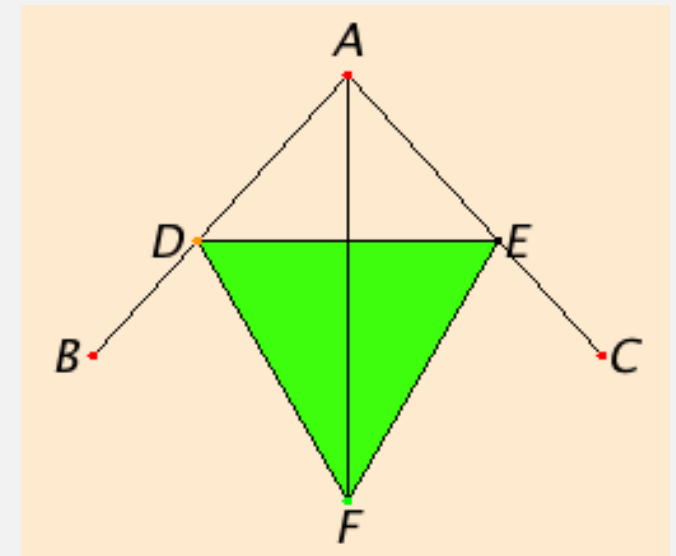
I say that the angle BAC is bisected by the straight line AF .

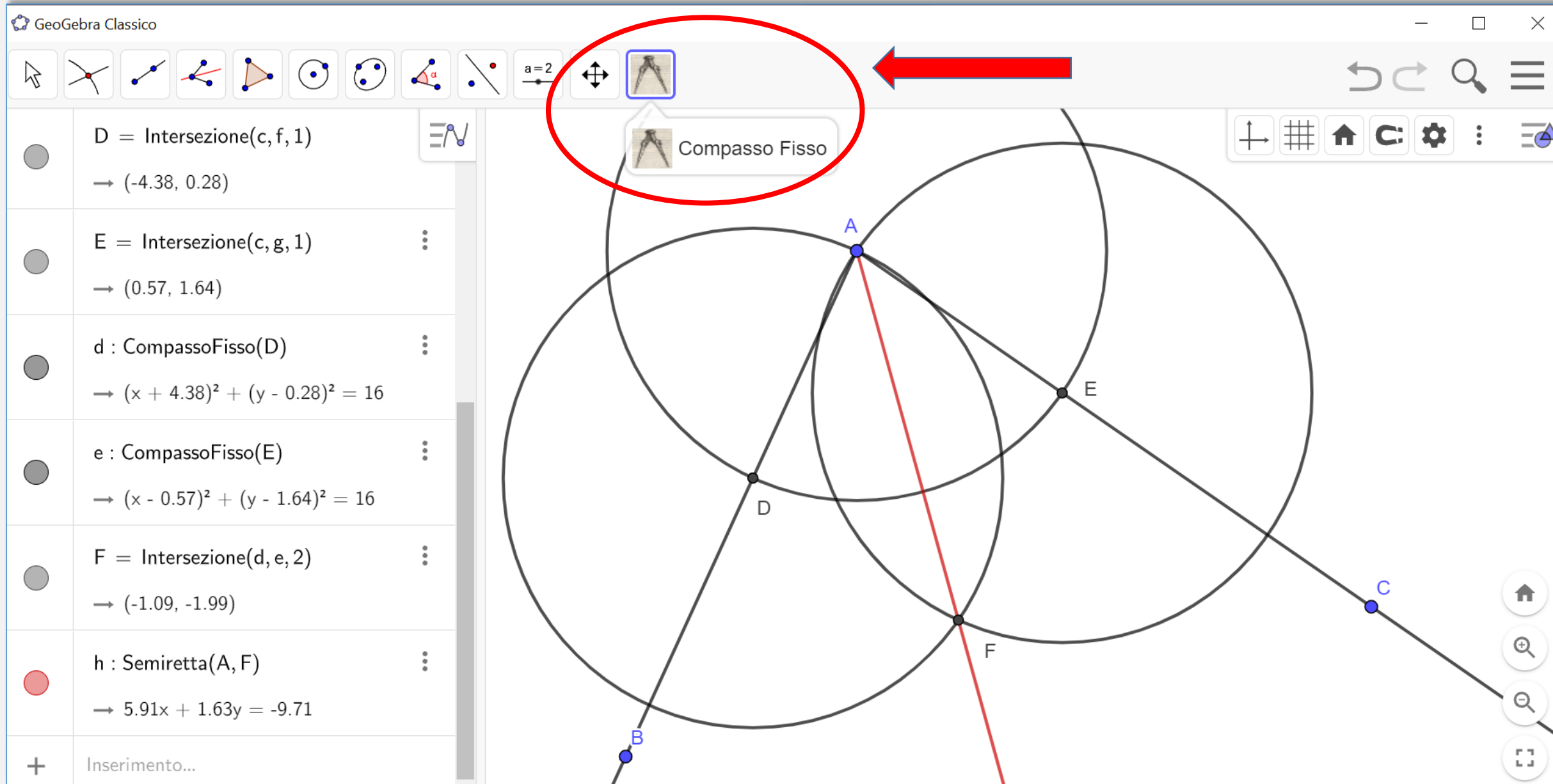
Since AD equals AE , and AF is common, therefore the two sides AD and AE equal the two sides EA and AF respectively [Definition 20]

And the base DF equals the base EF , therefore the angle DAF equals the angle EAF [Proposition I.8]

Therefore the given rectilinear angle BAC is bisected by the straight line AF .

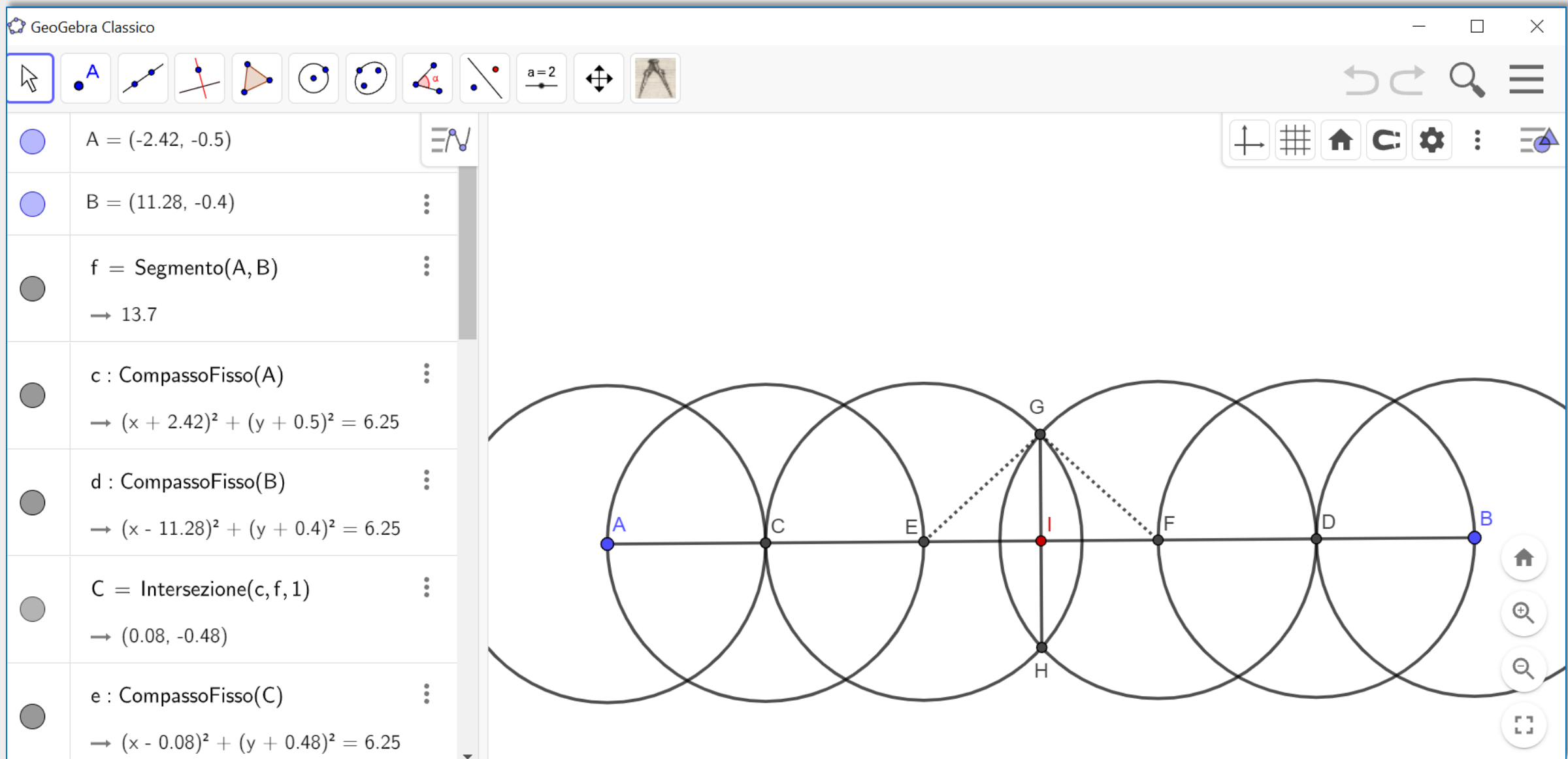
Ferrari	Euclide
F.1	I.4
F.2	I.5
F.3	I.8





F.4

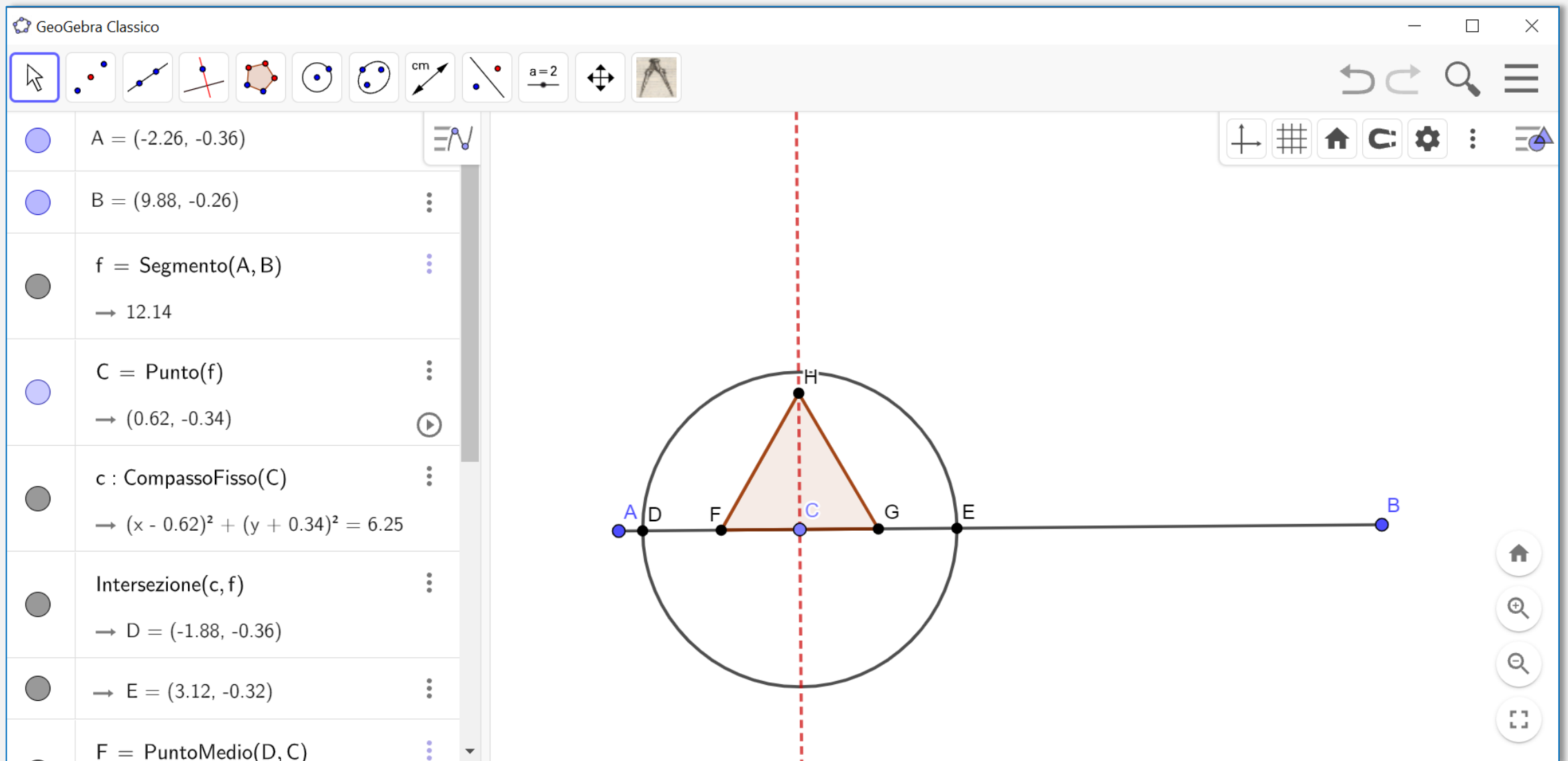
Elements I.9 (P) To bisect a given rectilinear angle



F.5

Elements I.10 (P)

To bisect a given finite straight line



F.6

Elementi I.11 (P)

To draw a straight line at right angles to a given straight line from a given point on it.

F.11		“Sopra una data linea, costruire un triangolo isoscele”.
F.12	<i>Elementi</i> I.2	“Condurre un segmento da un punto dato uguale ad un altro segmento assegnato”.
F.13	<i>Elementi</i> I.3	“Dati due segmenti disuguali, tagliare dal segmento maggiore una parte uguale al segmento minore”.
F.14	<i>Elementi</i> I.16	“In ogni triangolo un angolo esterno è sempre maggiore di tutti gli angoli interni del triangolo”.
F.15	<i>Elementi</i> I.17	“In ogni triangolo la somma di qualsiasi due angoli è minore di due angoli retti”.
F.16	<i>Elementi</i> I.18	“In ogni triangolo l’angolo maggiore è quello opposto al lato maggiore”.
F.17	<i>Elementi</i> I.19	“In ogni triangolo il lato maggiore è quello opposto all’angolo maggiore”.
F.18	<i>Elementi</i> I.20	“In ogni triangolo la somma di due lati è maggiore di quello rimanente”.
F.19	<i>Elementi</i> I.21	“Se internamente a un triangolo, da due vertici si tracciano due rette che si incontrano in un punto, formando un triangolo, la somma dei due lati di questo triangolo escludendo quello in comune è minore della somma dei due lati rimanenti del triangolo esterno”.
F.20	<i>Elementi</i> I.26	“Due triangoli sono uguali se hanno ordinatamente uguali un lato e i due angoli ad esso adiacenti”.
F.21	<i>Elementi</i> I.27	“Se due rette tagliate da un’altra retta formano angoli alterni uguali, allora sono parallele”.
F.22	<i>Elementi</i> I.28	“Se due rette tagliate da un’altra retta formano angoli corrispondenti uguali, allora sono parallele”.
F.23	<i>Elementi</i> I.29	“Se due rette parallele sono tagliate da un’altra retta allora formano angoli alterni uguali, angoli corrispondenti uguali e angoli coniugati la cui somma è di due angoli retti”.
F.24	<i>Elementi</i> I.30	“Rette parallele alla stessa retta sono parallele tra loro”.

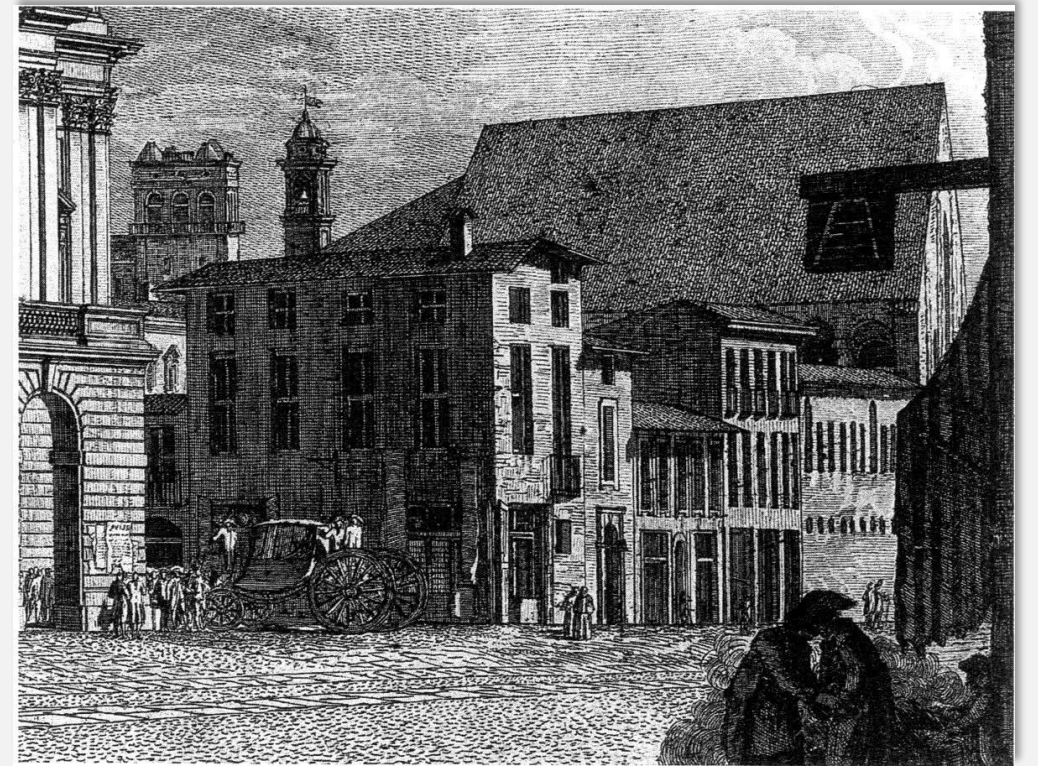
F.25	<i>Elementi</i> I.23	“Costruire su una retta data e con vertice in un dato punto di essa, un angolo rettilineo uguale a un angolo rettilineo dato”.
F.26	<i>Elementi</i> I.6	“Se un triangolo ha due angoli uguali, allora i lati opposti a questi angoli sono uguali”.
F.27	<i>Elementi</i> I.7	“Su una retta data e da ciascun suo estremo si conducano due rette che si incontrino in un punto; non è possibile costruire con gli stessi estremi e dalla stessa parte, altre due rette rispettivamente uguali a quelle prima costruite e aventi un diverso punto di incontro”.
F.28	<i>Elementi</i> I.24	“Dati due triangoli che hanno due lati uguali, se gli angoli compresi tra questi lati sono uno maggiore dell’altro allora anche i lati opposti a tali angoli sono uno maggiore dell’altro”.
F.29	<i>Elementi</i> I.25	“Se due triangoli hanno due lati uguali e le basi una maggiore dell’altra allora anche gli angoli opposti alle basi sono uno maggiore dell’altro”.
F.30	<i>Elementi</i> I.31	“Condurre la parallela ad una retta data passante per un punto esterno ad essa”.
F.31	<i>Elementi</i> I.32	“Ogni angolo esterno ad un triangolo è maggiore della somma dei due angoli interni non adiacenti ad esso e la somma degli angoli di un triangolo è uguale alla somma di due angoli retti”.
F.32	<i>Elementi</i> I.1	“Costruire un triangolo equilatero su un segmento dato”.
F.33	<i>Elementi</i> I.12	“Condurre la perpendicolare a una retta data da un punto esterno essa”.
F.34- F.51	<i>Elementi</i> I.33- I.49	

How did the challenge end?

10 August 1548, Milano, Santa Maria del Giardino Church

Ferrari sent 31 questions to Tartaglia, who replied by sending his solutions. Neither Ferrari nor Tartaglia was satisfied with the answers they received, so they agreed to a true 'mathematical duel' in Milan.

Tartaglia wrote that he arrived with only his brother, while Ferrari arrived with important figures who cheered him on wildly. Tartaglia recounted that he was intimidated by this situation and abandoned the duel.



The story does not end there, though...

In *Part Five*, Tartaglia proposed his solution.

TAVOLA DE I CAPI CONTENUTI nel Terzo libro .



L terzo libro ha due capi solamente. nel primo si dichiara quante siano le propositioni di ciascun libro di Euclide : quante di quelle siano problemi, da risolvere col compasso, & in qual modo si risolvino con ogni apritura di compasso proposta dallo auerliario . 2 car. 63

Nel secondo si dichiarano ventidue quesiti delli trentauno proposti all'auttore da Hieronimo Cardano medico Milanese & Ludouico Ferario in publica disputa l'anno. 1547. 2 car 85

LA QVINTA PARTE DEL
GENERAL TRATTATO
DE' NVMERI ET MISVRE,
DI NICOLO TARTAGLIA;
NELLA QVALE SI MOSTRA IL MODO DE ESSEQVIRE
CON IL COMPASSO, ET CON LA REGHA TVTTI LI
PROBLEMI GEOMETRICI DI EVCLIDE ET
DA ALTRI PHILOSOPHI,
ET CON MODI TVI ISTEDIENTI, E BREVI DI QUELLI
DATI DA ESSO EVCLIDE,
MATERIA NON MEN' VTILE CHE NECESSARIA A' GEO-
METRICI, DESIGNATORI, PERSPECTIVI, ARCHITET-
toni, Ingegneri, & Machinadori, & naturali, come Mathematici.
CON SVOI PRIVILEGII.

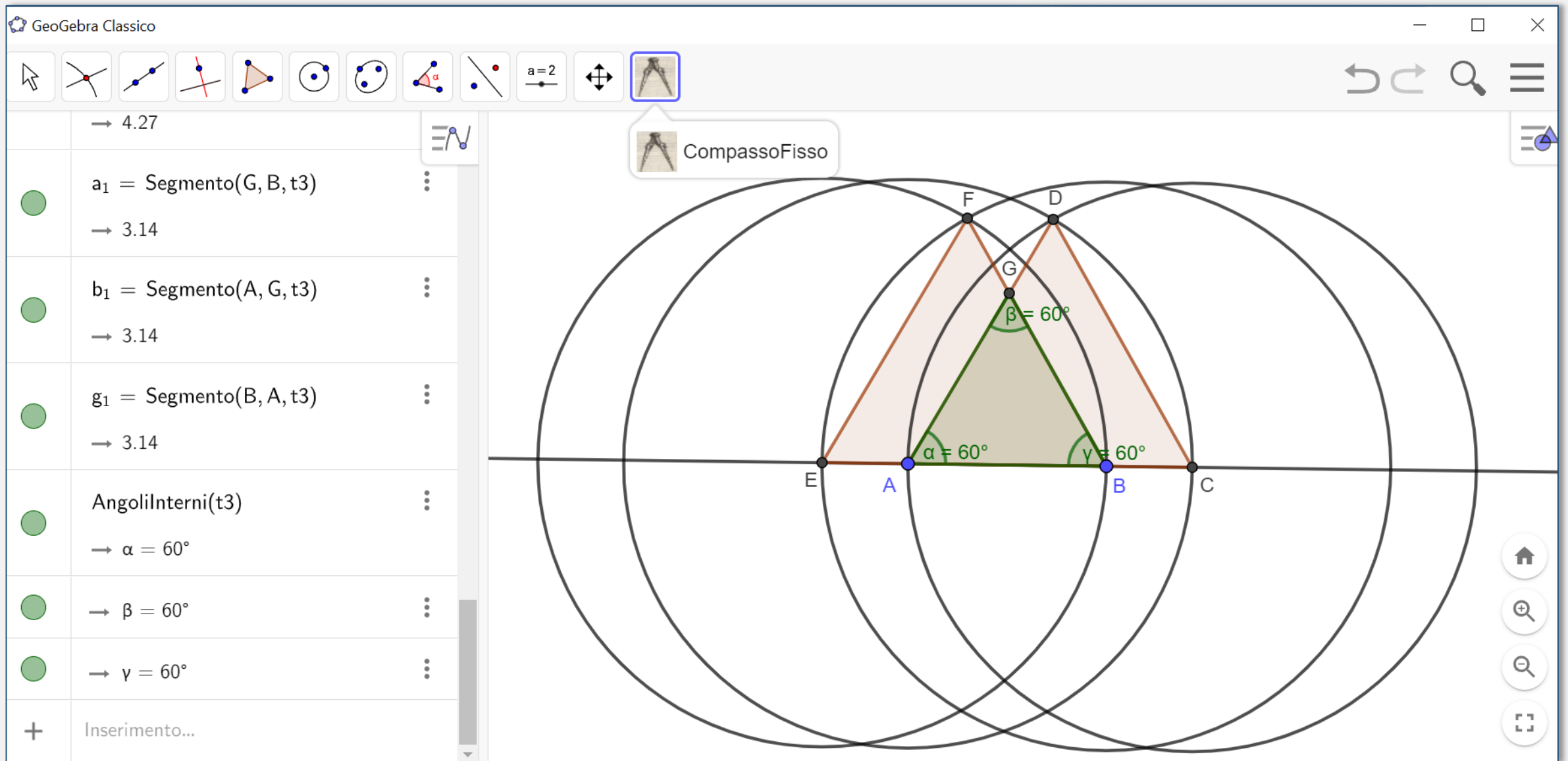


IN VENETIA PER CVRTIO TROIANO

LIBRO TERZO DELLA QVINTA
PARTE DEL GENERAL TRATTATO DI NVMERI,
ET MISVRE, DE NICOLO TARTAGLIA.

Which was the first proposition that Tartaglia proved?

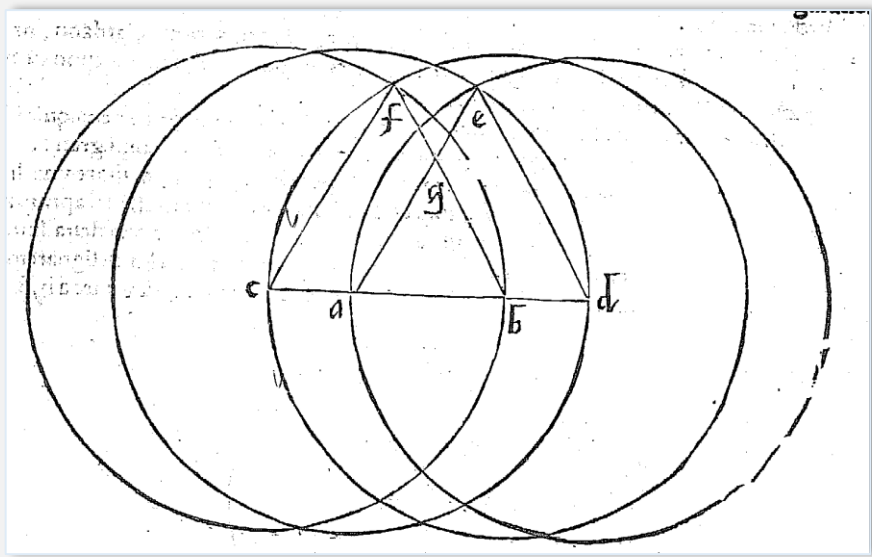
This is Proposition I.1 of the *Elements*, namely, the construction of an equilateral triangle **with a given side length**.



T.1

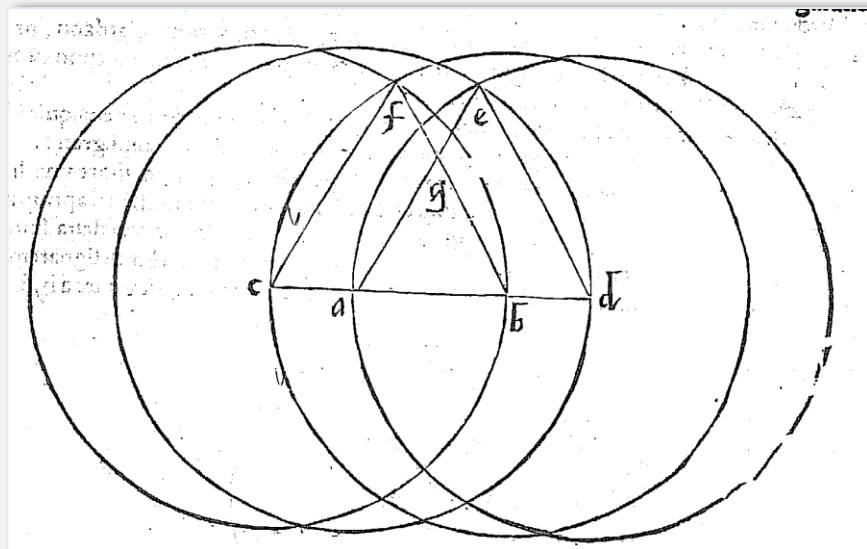
Elements I.1 (P)

To construct an equilateral triangle on a given finite straight line



We can construct an equilateral triangle with sides of any length determined by our opponent's compass opening. For example, suppose we want to construct an equilateral triangle with side length ab . I claim that we can construct such a triangle regardless of the fixed compass opening. If the compass opening were exactly ab , then by Euclid's Proposition I.1, we would follow that construction. However, if the fixed compass opening is greater than ab , then we draw two circles with centers at a and b , respectively.

We extend the line in both directions until it intersects the circles at points c and d , and we draw triangle aed with side AD , which is equilateral **by Euclid's Proposition I.1**. In fact, you can draw a circle with center d using the compass opening. We can do the same by taking point c as the center and drawing the equilateral triangle cfb . The sides fb and ea intersect at point g , and the triangle gab is equilateral because the two angles a and b are equal. In fact, they are the same angles as those of the equilateral triangles fb and aed . Therefore, **by Euclid's Proposition I.32 or VI.4**, the triangle gab is similar to the aforementioned equilateral triangles and is therefore equilateral.



What do you think of Tartaglia's approach to the problem of reconstructing the *Elements*?

appare poi allongaremo la detta linea da luna a l'altra parte per fina che seghi, ouer concorra alla circonferentia in li duoi ponti. c. & d. poi sopra la parte. a d. gli descriueremo, el triangolo a e d. equilatero per la dottrina della prima del primo di Euclide (cioe facendo vn altro cerchio sopra el centro, ouer ponto. d. secondo la sua apertura il medemo farai sopra la parte. c b. facendo il ponto. c. centro, & descriuendo, il detto triangolo equilatero. c f b. del quale el lato f b. se sega con el lato. e a. de l'altro triangolo in ponto. g. hor dico che el triangolo. g a b. è equilatero perche li duoi angoli. a. & b. (del detto triangolo. g a b.) sono eguali, anzi sono quelli istessi de li altri triāgoli equilateri. e f b. & a e d. onde (per la. 32. del primo di Euclide ouer p̄ la 4. del 6.) il triangolo. g a b. fara simile a cadauno de ditti duoi triangoli equilateri adonque è equilatero, che il proposito.

As we have seen, there are two approaches.

1. Ferrari attempts to construct an axiomatic-deductive system based on Euclid's postulates 1, 2, 4, and 5 and the modified third postulate.
2. Tartaglia considers “traditional” Euclidean geometry and combines it with Euclidean geometry incorporating the modified third postulate.

The questions we can ask students — and which I now pose to you — are as follows:

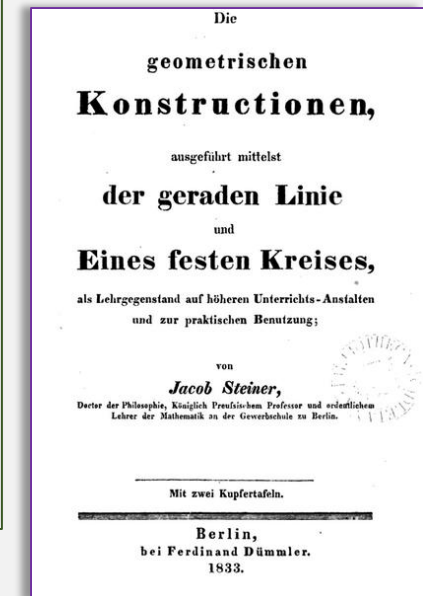
Can we say that one approach is more correct than the other? If so, which one? Why?

Before the *Cartelli*, the geometry of the straightedge and fixed compass appears to have consisted of a series of practical “rules”.

Ferrari, Cardano, Benedetti and Tartaglia, however, addressed the issue from a theoretical perspective.

1797: 'Any geometric construction that can be performed with a straightedge and compass can also be performed using only a compass' (Mascheroni, Mohr).

1833: Any geometric construction that can be performed using only a compass can also be performed using only a straightedge, provided that a fixed, complete circle is given on the sheet, in addition to the centre (Steiner-Poncelet).



Ludovico Ferrari's Theorem (1547)

Using a straightedge and a compass with a fixed opening, one can prove all theorems of plane geometry and perform all related constructions. However, circles with a radius different from the fixed opening cannot be drawn completely. One can, however, construct their center, radius, and as many points as desired.

(Amedeo Agostini, *Enciclopedia delle Matematiche Elementari*)

Thank you!!

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